

MISCELLANEOUS EXERCISE – CHAPTER 15 (MENSURATION)

(For Class 6 – with full explanations)

1. The perimeter of an isosceles triangle ABC shown in the adjoining figure is 82.8 cm. What is the length of AB?

In the figure, $BC = 22.4$ cm and $AB = AC$ (isosceles triangle).

Let $AB = AC = x$ cm.

$$\begin{aligned}\text{Perimeter} &= AB + BC + AC \\ \Rightarrow 82.8 &= x + 22.4 + x = 2x + 22.4\end{aligned}$$

$$\begin{aligned}\text{So,} \\ 2x &= 82.8 - 22.4 = 60.42\end{aligned}$$

$$x = 60.4 \div 2 = 30.2 \text{ cm}$$

Length of AB = 30.2 cm ✓

2. Find the area and perimeter of a square, if length of its side is :

Formulae for a square of side s :

$$\text{Perimeter } P = 4s$$

$$\text{Area } A = s^2$$

(i) 81 cm

$$P = 4 \times 81 = 324 \text{ cm}$$

$$A = 81 \times 81 = 6561 \text{ sq. cm}$$

(ii) 8.3 cm

$$P = 4 \times 8.3 = 33.2 \text{ cm}$$

$$A = 8.3 \times 8.3 = 68.89 \text{ sq. cm}$$

(iii) 4.7 cm

$$P = 4 \times 4.7 = 18.8 \text{ cm}$$

$$A = 4.7 \times 4.7 = 22.09 \text{ sq. cm}$$

3. Find the area and perimeter of a rectangle, if length and breadth of its sides are :

Formulae for a rectangle of length l, breadth b:

Perimeter $P=2(l+b)$

Area $A=l \times b$

(i) $l= 12 \text{ m}, b= 8 \text{ m}$

$$P=2(12+8)=2 \times 20=40 \text{ m}$$

$$A=12 \times 8= 96 \text{ sq. m}$$

(ii) $l= 10 \text{ m}, b= 6 \text{ m}$

$$P=2(10+6)= 2 \times 16=32 \text{ m}$$

$$A=10 \times 6= 60 \text{ sq. m}$$

(iii) $l= 4.5 \text{ cm}, b= 3.5 \text{ cm}$

$$P=2(4.5+3.5)= 2 \times 8=16 \text{ cm}$$

$$A= 4.5 \times 3.5$$

$$=4.5 \times (3+0.5)= 13.5+2.25=15.75 \text{ sq. cm}$$

4. What happens

(i) to the area of a rectangle, if its length is doubled, breadth remains the same?

Area of rectangle = length $\times$ breadth.

If length is doubled:

$$\text{New area} = (2 \times \text{length}) \times \text{breadth} = 2 \times (\text{length} \times \text{breadth}).$$

So **area becomes double.**

(ii) to the area of a rectangle, when its breadth is doubled, the length remains the same?

$$\text{New area} = \text{length} \times (2 \times \text{breadth})$$

$$= 2 \times (\text{length} \times \text{breadth}).$$

So **area becomes double.**

(iii) to the area of the rectangle, when its length as well as breadth are reduced to half?

New area

$$= (1/2 \text{ length}) \times (1/2 \text{ breadth})$$

$= \frac{1}{4} \times \text{original area.}$

So **area becomes one-fourth.**

(iv) to the area of a square, when its side is made half?

Area of square = side^2 .

New side = $\frac{1}{2} \times \text{original side.}$

New area

$= (\frac{1}{2} \text{ side})^2 = \frac{1}{4} \times \text{original area.}$

So **area becomes one-fourth.**

(v) to the area of a square, when its side is made double?

New side = $2 \times \text{original side.}$

New area

$= (2 \text{ side})^2 = 4 \times (\text{side})^2 = 4 \times \text{original area.}$

So **area becomes four times.**

(Exactly as in the answer key.)

5. A square carpet of dimensions $4 \text{ m} \times 4 \text{ m}$ is laid in a rectangular field of dimensions $8 \text{ m} \times 6 \text{ m}$. Find the area of the floor left uncovered.

Area of rectangular room

$= 8 \times 6 = 48 \text{ sq. m}$

Area of square carpet

$= 4 \times 4 = 16 \text{ sq. m}$

Uncovered area

$= \text{room area} - \text{carpet area}$

$= 48 - 16 = 32 \text{ sq. m}$

Answer: 32 sq. m ✓

6. Find the perimeter and area of the following shaded parts.

(i) First L-shaped figure

Divide the L-shape into two rectangles:

- Rectangle 1: $12 \text{ m} \times 4 \text{ m} \rightarrow \text{area} = 12 \times 4 = 48 \text{ sq. m}$
- Rectangle 2: $5 \text{ m} \times 5 \text{ m} \rightarrow \text{area} = 5 \times 5 = 25 \text{ sq. m}$

Total area
 $= 48 + 25 = 73 \text{ sq. m}$

Perimeter: add all outside sides.

Top = 12 m

Right vertical = 4 m

Inner horizontal = $12 - 5 = 7 \text{ m}$

Inner vertical = 5 m

Bottom = 5 m

Left vertical = $4 + 5 = 9 \text{ m}$

Perimeter

$= 12 + 4 + 7 + 5 + 5 + 9$

$= 42 \text{ m}$

Answer: P = 42 m, A = 73 sq. m ✓

(ii) Second L-shaped figure

Think of a big rectangle of dimensions $10 \text{ m} \times 6 \text{ m}$, from which a small rectangle $4 \text{ m} \times 3 \text{ m}$ is removed.

Area of big rectangle

$= 10 \times 6 = 60 \text{ sq. m}$

Area of small cut-out rectangle

$= 4 \times 3 = 12 \text{ sq. m}$

Area of L-shape

$= 60 - 12 = 48 \text{ sq. m}$

Perimeter: follow the boundary.

Bottom = 10 m

Right vertical = 3 m

Inner horizontal = 4 m

Inner vertical = 3 m

Top = 6 m

Left vertical = 6 m

Perimeter

$= 10 + 3 + 4 + 3 + 6 + 6$

$= 32 \text{ m}$

Answer: P = 32 m, A = 48 sq. m ✓

(iii) Shaded border of a square frame

Outer square side = 20 cm

Inner square side = 18 cm

Area of outer square = $20 \times 20 = 400$ sq. cm

Area of inner square = $18 \times 18 = 324$ sq. cm

Area of shaded region

= $400 - 324 = 76$ sq. cm

Perimeter of shaded region includes **both** outer and inner boundaries.

Outer perimeter = $4 \times 20 = 80$ cm

Inner perimeter = $4 \times 18 = 72$ cm

Total perimeter

= $80 + 72 = 152$ cm

Answer: P = 152 cm, A = 76 sq. cm ✓

(iv) Cross-shaped figure

From the figure:

- Total width = $10 + 2 + 10 = 22$ cm
- Total height = $6 + 2 + 6 = 14$ cm

Consider a big rectangle $22 \text{ cm} \times 14 \text{ cm}$ and remove four corner rectangles each $10 \text{ cm} \times 6 \text{ cm}$.

Area of big rectangle

= $22 \times 14 = 308$ sq. cm

Area of one corner rectangle

= $10 \times 6 = 60$ sq. cm

Area removed = $4 \times 60 = 240$ sq. cm

Area of cross

= $308 - 240 = 68$ sq. cm

Perimeter of cross: it equals the perimeter of the outer $22 \text{ cm} \times 14 \text{ cm}$ rectangle.

Perimeter

= $2(22 + 14) = 2 \times 36 = 72$ cm

Answer: P = 72 cm, A = 68 sq. cm ✓

7. Find the surface area and volume of a cube, length of whose edge is :

For a cube of side a :

$$\text{Surface area } S = 6a^2$$

$$\text{Volume } V = a^3$$

(i) 6 cm

$$S = 6 \times 6^2 = 6 \times 36 = 216 \text{ sq. cm}$$

$$V = 6^3 = 6 \times 6 \times 6 = 216 \text{ cubic cm}$$

(ii) 6.5 cm

$$\text{First } 6.5^2 = 42.25.$$

Surface area

$$S = 6 \times 42.25 = 253.5 \text{ sq. cm}$$

Volume

$$V = 6.5^3 = 6.5 \times 42.25 = 274.625 \text{ cubic cm}$$

(iii) 8 cm

$$S = 6 \times 8^2 = 6 \times 64 = 384 \text{ sq. cm}$$

$$V = 8^3 = 8 \times 8 \times 8 = 512 \text{ cubic cm}$$

8. Find the surface area and volume of a cuboid whose length, breadth and height are given as :

For cuboid l, b, h :

$$\text{Surface area } S = 2(lb + bh + hl)$$

$$\text{Volume } V = l \times b \times h$$

(i) $l = 8 \text{ cm}$, $b = 8 \text{ cm}$, $h = 6 \text{ cm}$

$$lb = 8 \times 8 = 64$$

$$bh = 8 \times 6 = 48$$

$$hl = 6 \times 8 = 48$$

$$\text{Sum} = 64 + 48 + 48 = 160$$

Surface area

$$S = 2 \times 160 = 320 \text{ sq. cm}$$

Volume

$$V = 8 \times 8 \times 6 = 384 \text{ cubic cm}$$

(ii) $l = 8.3 \text{ m}$, $b = 4.7 \text{ m}$, $h = 2.3 \text{ m}$

$$lb = 8.3 \times 4.7 = 39.01$$

$$bh = 4.7 \times 2.3 = 10.81$$

$$hl = 2.3 \times 8.3 = 19.09$$

$$\text{Sum} = 39.01 + 10.81 + 19.09 = 68.91$$

Surface area

$$S = 2 \times 68.91 = 137.82 \text{ sq. m}$$

Volume

$$V = 8.3 \times 4.7 \times 2.3$$

$$\text{First } 8.3 \times 4.7 = 39.01$$

$$\text{Then } 39.01 \times 2.3 = 78.02 + 11.703 = 89.723 \text{ cubic m}$$

Answers:

(i) $S = 320 \text{ sq. cm}$, $V = 384 \text{ cubic cm}$

(ii) $S = 137.82 \text{ sq. m}$, $V = 89.723 \text{ cubic m}$ ✓

9. The volume of a cuboid is 3600 cubic m. If its length is 20 m and breadth is 15 m, find its height.

Let height = $h \text{ m}$.

$$\text{Volume } V = l \times b \times h$$

So,

$$3600 = 20 \times 15 \times h$$

$$20 \times 15 = 300$$

$$3600 = 300 h$$

$$h = 3600 \div 300 = 12 \text{ m}$$

Answer: Height = 12 m ✓

10. How many litres of water can a water tank hold if its length is 4.5 m, breadth is 3 m and height is 3 m?

Volume of tank in cubic metres
 $V = 4.5 \times 3 \times 3 = 4.5 \times 9 = 40.5$ cubic m

1 cubic metre = 1000 litres

Capacity in litres
 $= 40.5 \times 1000 = 40500$ litres

Answer: 40500 litres ✓

ASSERTION AND REASON (Page 262)

Choose:

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

1.

Assertion (A): If a square, a rectangle and a pentagon are all made from a string of length 32 cm, then the perimeter of each figure will be different.

Reason (R): Perimeter is the length of the boundary of a closed figure.

- The same string of length 32 cm will give **perimeter 32 cm for each figure**, so the assertion is **false**.
- The reason correctly defines perimeter, so it is **true**.

Correct option: (d) – A is false, but R is true. ✓

2.

Assertion (A): The perimeter of square whose area is 36 cm^2 is 24 cm.

Reason (R): Area of the square = $6 \text{ cm} \times 6 \text{ cm}$ and Perimeter of the square = $4 \times 6 \text{ cm} = 24 \text{ cm}$.

- If area is 36 cm^2 , side = $\sqrt{36} = 6 \text{ cm}$, so perimeter = $4 \times 6 = 24 \text{ cm}$. Assertion is **true**.
- The reason shows the full calculation and correctly explains why the perimeter is 24 cm. Reason is **true** and explains A.

Correct option: (a) – Both A and R are true and R is the correct explanation. ✓

3.

Assertion (A): If the side of a square is doubled, the perimeter of the square is halved.

Reason (R): Perimeter of the square = $4 \times \text{side}$.

- If side is doubled, perimeter becomes $4 \times (2 \times \text{side}) = 8 \times \text{side}$, which is **double**, not half. So assertion is **false**.
- The formula for perimeter is correct. Reason is **true**.

Correct option: (d) – A is false, but R is true. ✓

4.

Assertion (A): The length of a rectangular field of area 240 m^2 and width 12 m is 2000 cm.

Reason (R): Area of the rectangle = length + breadth.

- Area = length $\times$ breadth.
So length = area $\div$ breadth = $240 \div 12 = 20 \text{ m}$.
 $20 \text{ m} = 20 \times 100 = 2000 \text{ cm}$. Thus assertion is **true**.
- The reason states area = length + breadth, which is **wrong**.

Correct option: (c) – A is true, but R is false. ✓